

Solutions

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1 Overview

This document handles all **R-side preprocessing**: reading raw inputs from `data/`, cleaning, and writing tidy outputs that `computation.qmd` consumes.

Project root is resolved with `here::here()` so paths work regardless of the render working directory.

```
data_dir <- here("prod-ftn", "data")
list.files(data_dir)
```

```
#> character(0)
```

2 Load

```
# Example: read every CSV in data/ into a named list of data.tables.
files <- list.files(data_dir, pattern = "\\\\.csv$", full.names = TRUE)
raw <- setNames(lapply(files, fread), tools::file_path_sans_ext(basename(files)))
```

Table 1: Summary of cleaned data

```
tidy |>
  summarise(across(where(is.numeric), list(mean = mean, sd = sd), .names = "{.col}_{.fn}")) |>
  kbl(booktabs = TRUE, digits = 3) |>
  kable_styling(latex_options = "hold_position")
```

3 Clean

```
# Example transform - replace with real cleaning logic.
tidy <- raw$main |>
  as_tibble() |>
  filter(!is.na(id)) |>
  mutate(date = ymd(date)) |>
  arrange(id, date)
```

4 Summarize

5 Export

```
# Outputs for the Julia computation step.
fwrite(tidy, file.path(data_dir, "clean.csv"))
```

6 Math

$$y_{it} = \beta \text{SAM}_{it} + \varepsilon_{it}. \quad (1)$$